

JNEOPALLIUM · INDUSTRIAL LOOP GUARDIAN

Your Plant Already Sends the Signals. Start Reading Them.

Cross-loop, cross-timescale operational intelligence from your existing OPC UA & MQTT telemetry

On-premises · Shadow → Advisory · Supervisory above PLC/PID/SIS · BSD-3 core

01 Your PID holds the variable. The plant still loses money.

The expensive problems are the ones no single loop can see.

- ▶ A pump wears out over **weeks** — the "bearing > 85 °C" alarm fires only after the damage.
- ▶ Three controllers each behave correctly while together **wasting energy** no loop detects.
- ▶ The same rising temperature is normal at startup and alarming at steady state — **nuisance alarms**.
- ▶ Is the process hotter, or is one sensor drifting? Majority voting **can't tell**.
- ▶ Each of these is a decision across **several signals on different timescales** — not a single loop.

02 We supervise. We don't replace.

Keep — unchanged

- PLC / PID / SIS deterministic control
- Hard interlocks and emergency shutdown
- Your existing OPC UA & MQTT telemetry
- Operator overrides, authoritative

Add — Loop Guardian (above)

- Multi-loop, multi-timescale diagnosis
- Energy & bounded setpoint optimisation
- Degradation prediction & maintenance planning
- Advisories with an ****economic basis****

03 It watches across timescales

A single neuron weighs the instant against the trend.

- ▶ **Every tick:** flow, pressure, interlock state, pump command vs. actual speed — immediate condition.
- ▶ **Every 5–10 ticks:** vibration RMS — mechanical anomaly.
- ▶ **Every 10 ticks:** bearing temperature, power — thermal condition and efficiency.
- ▶ **Every 50+ ticks:** production load, degradation, maintenance schedule — operating and planning context.
- ▶ A PID sees one variable now. The Loop Guardian sees the **whole asset over time**.

04 Four detectors that pay for themselves

Narrow, high-value, and already trained in the demo.

- ▶ **Pump wear & cavitation risk** — distinguishes real degradation from a temporary high load.
- ▶ **Control-loop oscillation & tuning degradation** — separates valve stiction from normal transients.
- ▶ **Temperature-sensor drift** — tells a drifting sensor from a genuine process rise, avoiding a needless trip.
- ▶ **Energy-per-unit-production deterioration** — flags a loop that holds its variable while wasting energy.

05 Now it also hears and feels your machines

New: a multimodal machine-health subsystem — acoustic + vibration condition monitoring.

- ▶ **Listens and feels** — turns sound and vibration into a calibrated health score and named faults.
- ▶ **Names the fault** — bearing damage, cavitation, imbalance, sensor fault, or an unknown anomaly.
- ▶ **Knows when it's unsure** — a domain-shift score and an uncertainty value, so an unfamiliar machine reads as "not sure," not a false alarm.
- ▶ **Learns from public benchmarks** — MIMII, DCASE, and the Paderborn bearing dataset, plus your own telemetry once reviewed.
- ▶ **Read-only by construction** — it advises; it never touches an actuator.

06 Every advisory carries its economic basis

Not just "there's a problem" — what it is, why, and what it's worth.

- ▶ `finding`: pump wear and cavitation risk · `confidence`: 0.73
- ▶ `evidence`: PumpHealthAndEfficiencyNeuron, max vibration 4.3 mm/s
- ▶ `recommendation`: SCHEDULE_PUMP_INSPECTION · `urgencyHours`: 48
- ▶ `economicBasis`: estimated avoided-shutdown value **\$20,000**, safety envelope satisfied
- ▶ `autonomousAction`: **false** — and the PLC/PID/SIS boundary is stated in every record.

07 The profitability is simple to compute

Annual value $V = \text{Energy} + \text{Downtime} + \text{Quality} + \text{Maintenance} + \text{Operator} - \text{Cost}$. Illustrative example:

- ▶ Energy: 2% of an \$800k bill → **\$16,000**
- ▶ Avoided shutdowns: $2 \times \$20,000$ → **\$40,000**
- ▶ Reduced maintenance / scrap → **\$15,000**
- ▶ Gross annual value → **\$71,000** · platform & support → $-\$20,000$
- ▶ **Net annual value $\approx \$51,000$** . Price the product at ~10–20% of verified savings.

08 Proof: the pipeline works

Near-perfect separation on the reference skid — machinery proven, ready for your data.

0.9995

Macro-F1 across
the four detectors

~0.0005

Macro false-
positive rate

1.0

Fast-control availability
through an MQTT outage

0.1 s

Hard interlock latency
(owned by PLC/SIS)

09 Safe by construction

Easy to demonstrate, validate, insure, and sell.

- ▶ **Advisory by default.** It recommends; PLC/PID/SIS and operators decide. It never trips the plant.
- ▶ **Deterministic controls stay authoritative** — interlocks and overrides outrank any suggestion.
- ▶ **MQTT cannot actuate** (AUTONOMOUS is structurally rejected); only a bounded OPC UA path can command.
- ▶ **Fail-safe on loss:** lose the supervisor and fast control stays up; lose OPC UA and local fail-safe applies.
- ▶ It is **simulation/HIL evidence** today — not a certified SIS, and it never claims to be.

10 Land and expand — paid at every step

A low-risk path from a data export to a recurring subscription. (Indicative pricing.)

- ▶ **Phase A — Offline replay.** Analyse a historian export; deliver anomalies, timeline, and €-impact. **€3–8k / assessment.**
- ▶ **Phase B — Shadow pilot.** Read-only, no writes, 6–12 weeks; compare against real outcomes. **€7.5–20k**, credited to year one.
- ▶ **Phase C — Production advisory subscription.** Operator-visible advisories, one edge deployment. **€15–40k / site / year.**
- ▶ **Phase D — Bounded autonomous optimisation.** Only after evidence + a site safety case. **€30–100k+ / site / year.**

11 Where we don't compete — and where we do

Keep your PID (low/no value)

- A single stable regulatory loop
- Hard interlocks & emergency shutdown
- "pressure > limit → shut valve"
- Cheap downtime, little telemetry

Add Loop Guardian (high value)

- Predictive maintenance
- Oscillation & sensor-fault diagnosis
- Energy / setpoint optimisation
- Batch-context anomaly detection

12 Built to fit the plant you already run

On-premises, open core, integrator-friendly.

- ▶ **Reads what you already produce** — OPC UA process tags and MQTT IIoT telemetry; optional Kafka, CMMS, energy meters.
- ▶ **On-premises edge deployment** — local, HTTP cluster, or gRPC; no cloud required.
- ▶ **Open BSD-3 core** — no lock-in; the moat is the model packs, integrations, and support, not the engine.
- ▶ **Integrator channel** — you bring the plant relationship; we bring the supervisory intelligence and releases.
- ▶ **Train-to-deploy in one step** — what you train is the exact network you run; every weight is inspectable.

Send us your data. We'll show you the money.

Export an anonymised historian sample, or point us at a simulated OPC UA endpoint.

We return discovered anomalies, an event timeline, an estimated €-impact, and a false-positive review.

Open-source demo today · paid assessment in weeks · shadow pilot next quarter.

Book a paid operational assessment.